

Avionics Navigation Systems

ESE482\ECE4635 - Fall 2026

Lecture 3

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Lecture Topic

The Navigation Equations

Chapter #2

Index of Chapter 2

1. Introduction
2. Geometry of the Earth
3. Coordinate Frames
4. Dead-Reckoning
5. Positioning
6. Course Computation
7. Navigation Errors

Introduction

How do we mathematically calculate the aircraft's position, velocity, heading, and route from navigation sensor measurements?

By Navigation equations

The navigation computer receives measurements from different sensors and processes them mathematically to determine the aircraft's navigation state.

Introduction

- Navigation equations process sensor data in the onboard computer to determine the aircraft's **position, velocity, and attitude**. They are part of the airborne software, with fixed instructions stored in **ROM** and mission-specific data, such as **waypoints**, stored in **RAM.**

- The aircraft navigation system uses three main types of sensor information:

1. **Absolute position data** from radio aids, radar, and satellites.
2. **Dead-reckoning data** from inertial, Doppler, and air-data sensors to estimate the current position.
3. **Star line-of-sight measurements** to determine position and attitude errors.

✓ **In short:** The navigation computer combines data from different sensors to continuously calculate and update the aircraft's position, velocity, and attitude.

Introduction

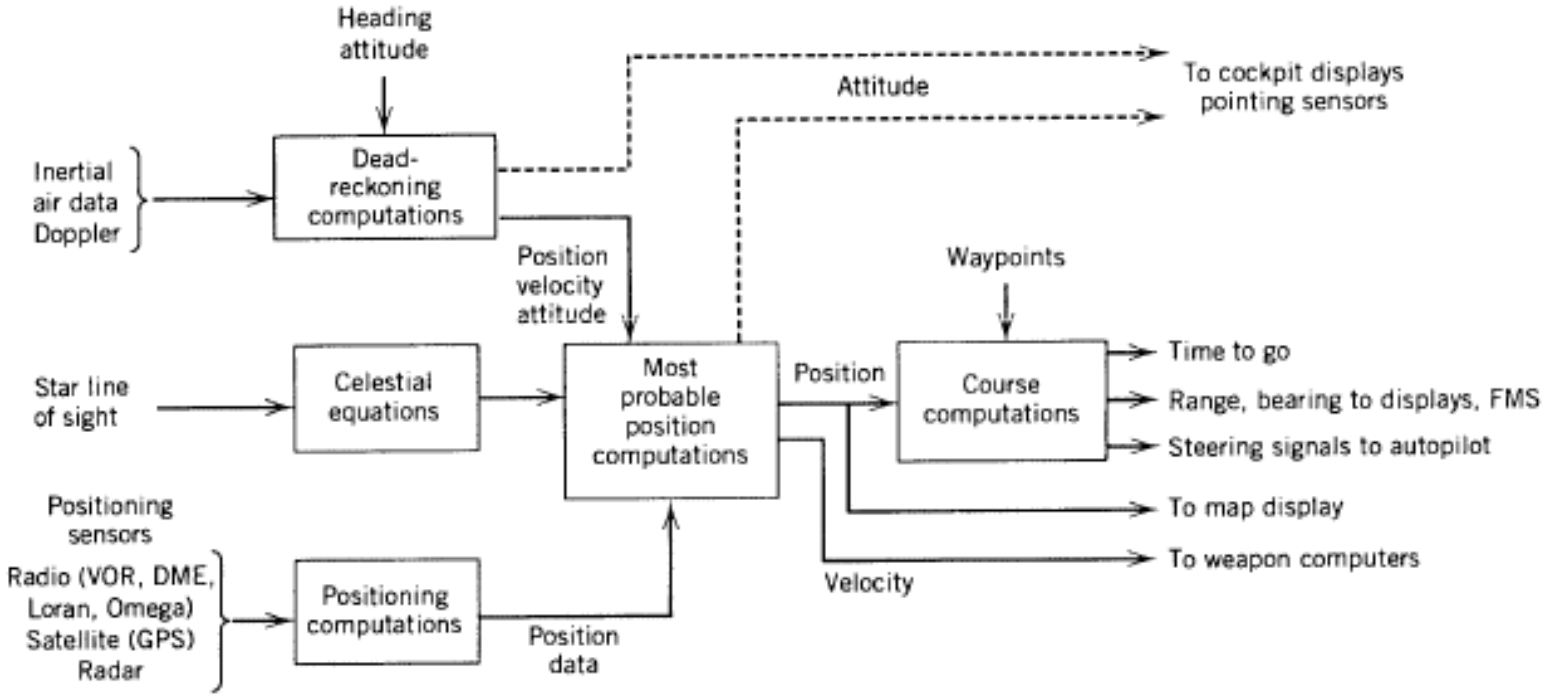
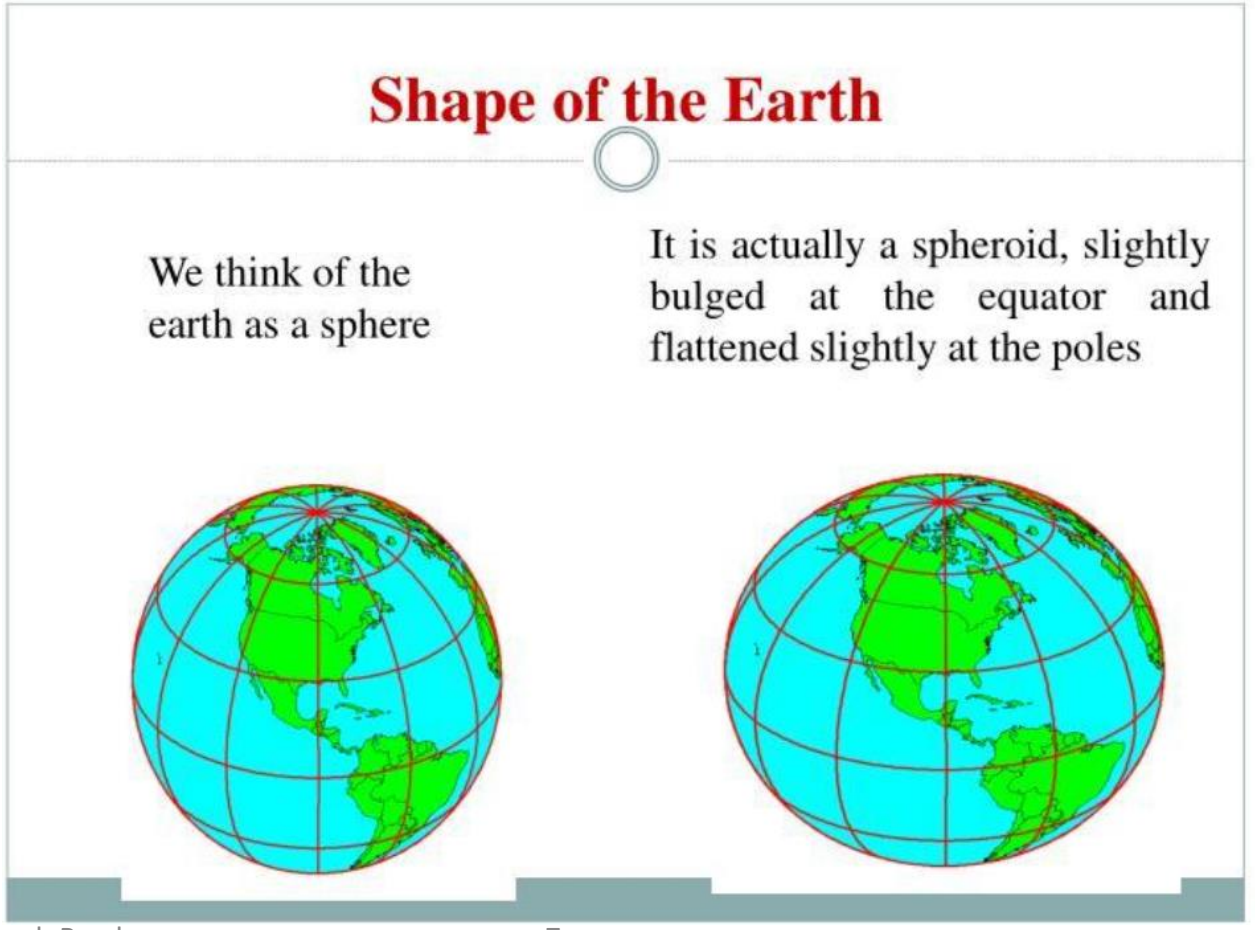


Figure 2.1 Block diagram of an aircraft navigation system.

Geometry of the Earth

➢ Why do we need Earth geometry?

- Because an aircraft does not move over a **flat Earth**.
- The Earth is approximately an **ellipsoid**.
- This means that if we want to accurately calculate the aircraft's **position, distance, or direction**, we need to take the **shape of the Earth** into account.



Gravity and Earth Rotation

➢ Earth's apparent gravity field

Simply, the gravity **we actually feel or measure at the Earth's surface** is not caused by gravitational force alone, because the Earth is rotating about its own axis.

➢ There are two effects involved:

1. **Gravitational force** → the force exerted by the Earth that attracts objects toward its center.
2. **Effect of Earth's rotation** → the rotation of the Earth produces an additional effect that makes the **apparent gravity** slightly less than the gravitational force alone.

✓ Therefore: **Apparent gravity = gravitational effect + effect due to Earth's rotation**

- **G** = gravitational field
- **Ω** = angular velocity of Earth
- **R** = position vector from Earth's center
- **g** = apparent gravity

$$\mathbf{g} = \mathbf{G} - \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{R})$$

➢ **Why Is This Important in Navigation?**

This is important because **Inertial Navigation Systems (INS)** rely on accelerometers to measure the aircraft's motion.

The accelerometers are affected by both **Earth's gravity** and **Earth's rotation**.

Therefore, the navigation computer must take **Earth's gravity and rotation** into account in order to accurately calculate the aircraft's **position and velocity**.

✓ "The apparent gravity of the Earth is the gravity actually experienced by a vehicle after accounting for the effect of Earth's rotation."

Earth as an Ellipsoid

Instead of considering the Earth to be a **perfect sphere**, navigation systems model it as an **ellipsoid**.

➢ **Why?**

Because the Earth is not perfectly spherical due to its rotation.

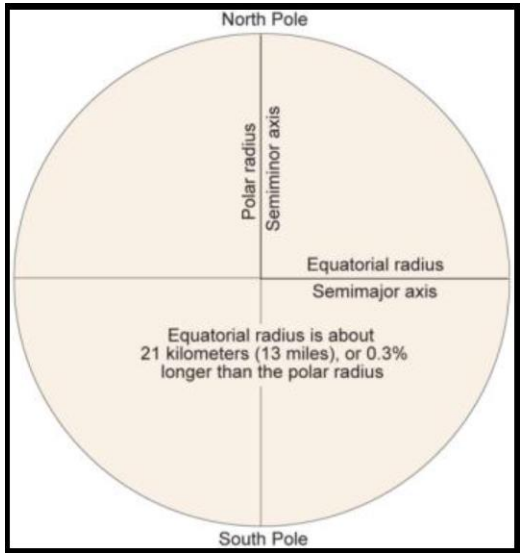
➢ **The ellipsoid is mainly characterized by:**

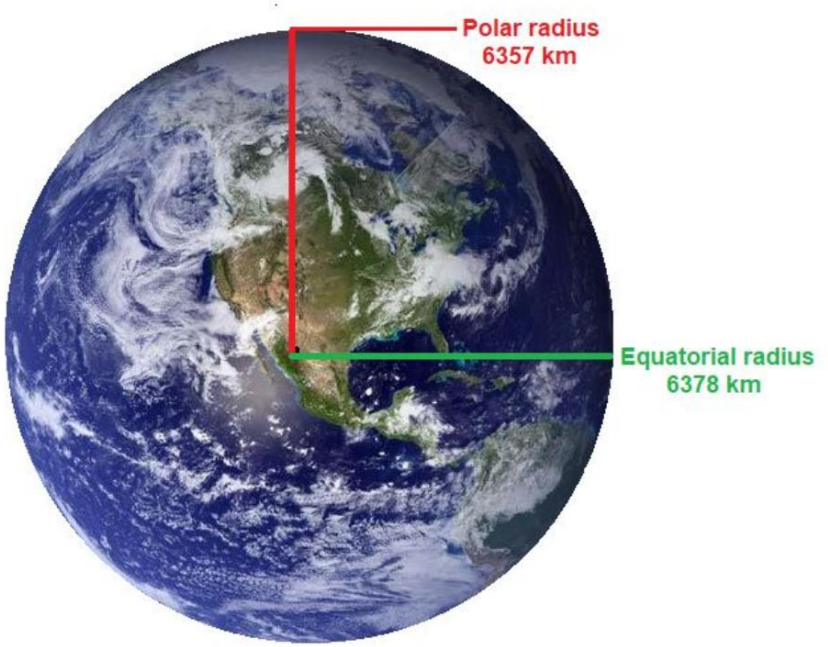
- Semi-major axis a (Equatorial radius)
- Semi-minor axis b (polar radius)

e: Eccentricity

$$e = \frac{\sqrt{a^2 - b^2}}{a}$$

✓ The ellipsoid used for navigation is selected to provide the **best fit to the actual shape of the Earth**.





Earth as an Ellipsoid

➢ **Flattening** refers to the **degree to which the Earth is flattened at the poles**.

The Earth is not a **perfect sphere**; it is slightly flattened at the poles and wider around the **equator**.

- a = semi-major axis
- b= semi-minor axis

$$f = \frac{a - b}{a}$$



➢ If the Earth were a perfect sphere: a = b Therefore: f = 0

However, because the Earth is **flattened at the poles**: a > b Therefore: f > 0

✈ Why Do We Need It in Navigation?

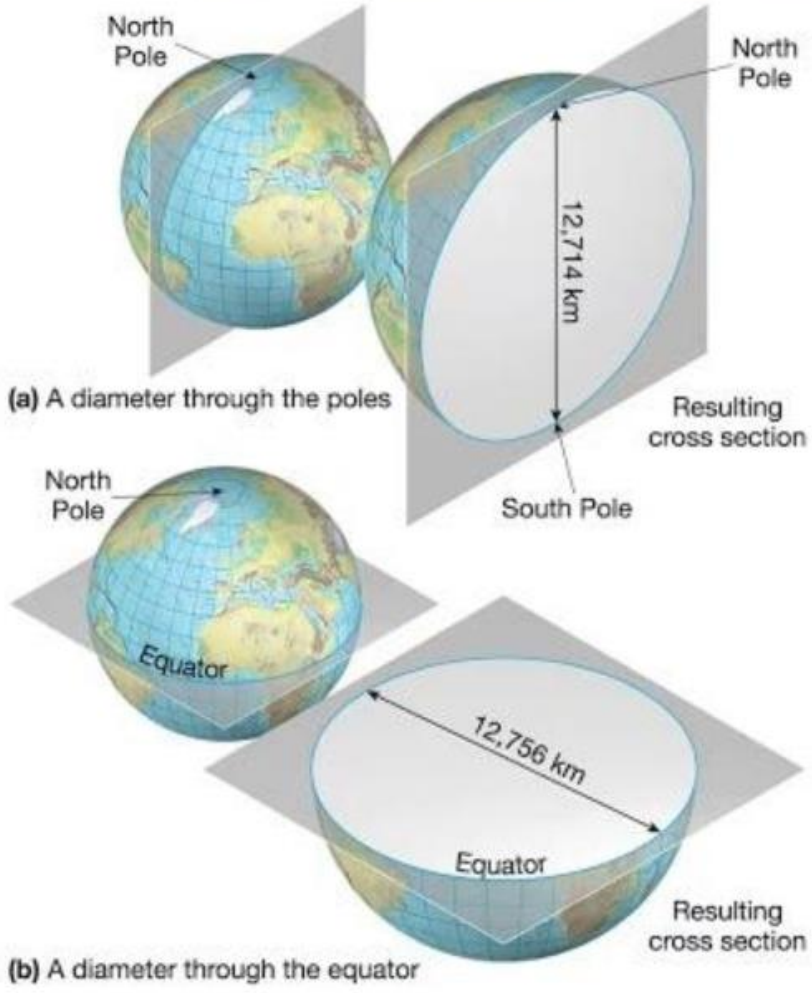
Accurate calculations of **aircraft position, distance, latitude, and longitude** require a realistic model of the Earth's shape.

Therefore, navigation systems use an **ellipsoid** rather than a perfect sphere.

✓ **"Flattening describes how much the Earth is compressed at the poles compared with a perfect sphere."**

Earth as an Ellipsoid

Earth Shape: an oblate spheroid



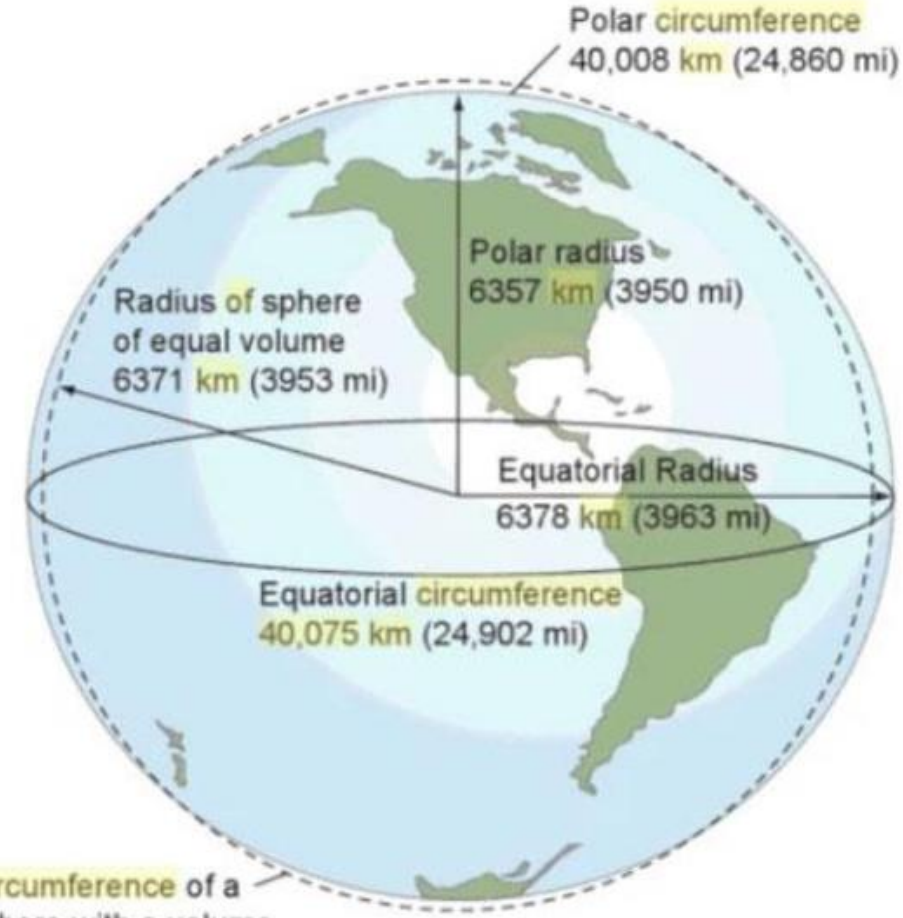


Figure 2.11 The true shape of the Earth geoid. The polar radius is 21 kilometers less than the equatorial radius, making Earth slightly less than a perfect sphere.

Latitude

We need to distinguish between the different types of **latitude** used in navigation.

- **Geodetic Latitude (φ_T)**

This is the **latitude commonly used in navigation**.

It is the angle between the normal to the reference ellipsoid and the equatorial plane.

- **Geocentric Latitude (φ_c)**

It is the angle between the equatorial plane and a line drawn from the Earth's center (R radius vector) to a point on its surface.

✓ The two latitudes are **not necessarily equal** because the Earth is an **ellipsoid**, not a perfect sphere.

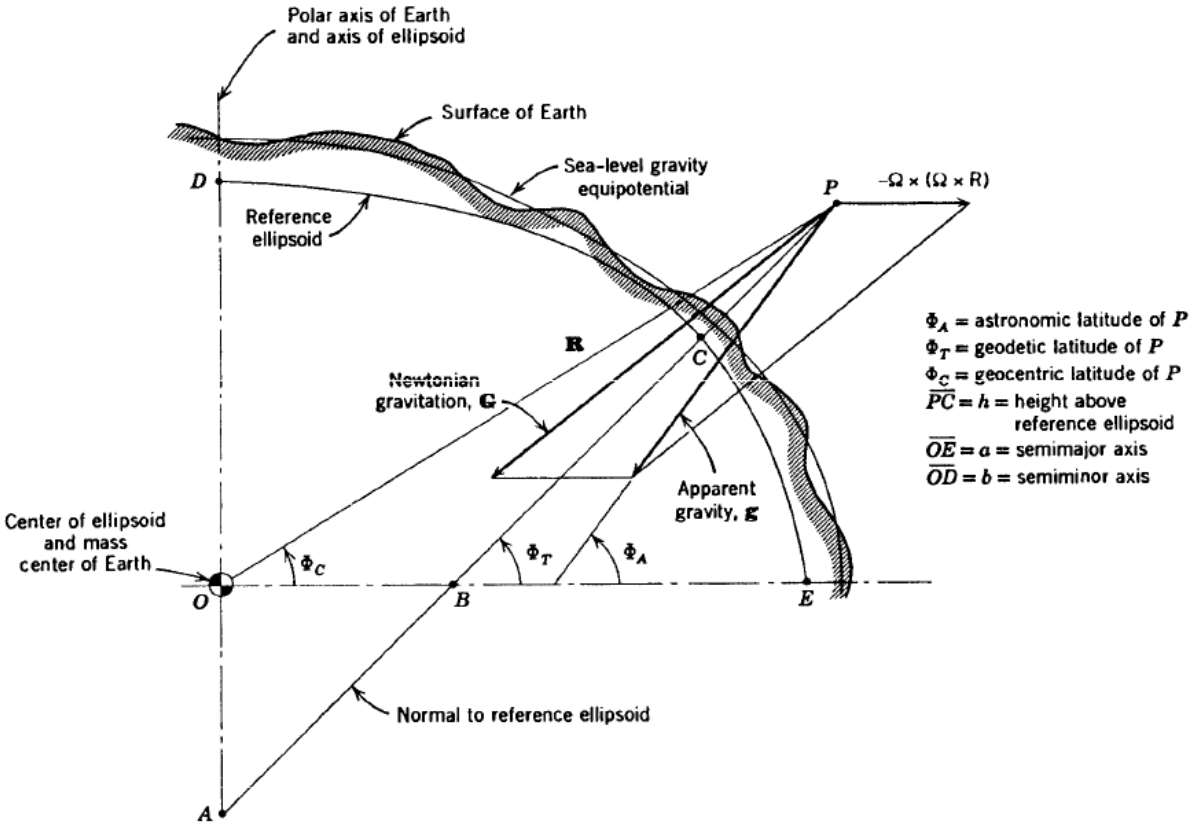
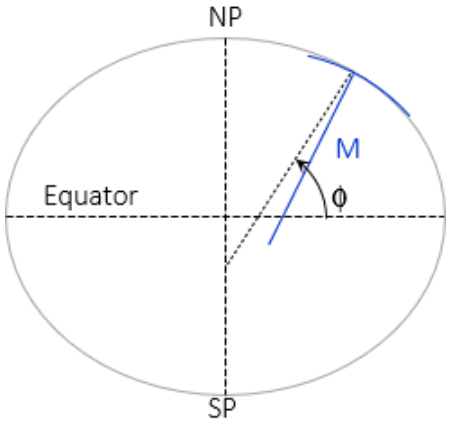


Figure 2.2 Meridian section of the Earth, showing the reference ellipsoid and gravity field.

Radius of Curvature

Because the Earth is curved, we need to know the radius of curvature at a particular location.
The radii of curvature of the ellipsoid are of fundamental importance to dead-reckoning navigation.



- **Meridian Radius of Curvature R_M**

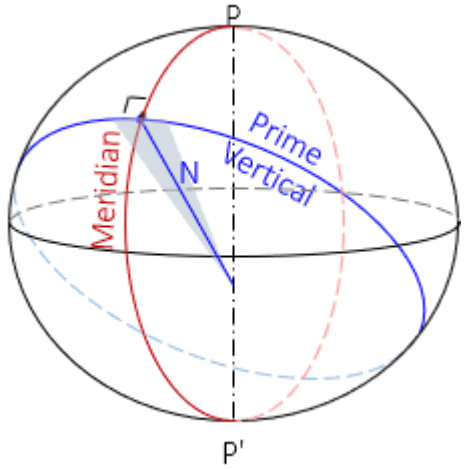
It is the radius of the best fitting circle to a meridian section of the ellipsoid
This is associated with motion in the **North-South direction**.

$$R_M = \frac{a(1-e^2)}{(1-e^2\sin^2\varphi_T)^{3/2}} \approx a\left[1+e^2\left(\frac{3}{2}\sin^2\varphi_T - 1\right)\right]$$

- **Prime Vertical Radius of Curvature R_P**

It is the radius of the best fitting circle to a vertical east-west section of the ellipsoid
This is associated with motion in the **East-West direction**.

$$R_P = \frac{a}{(1-e^2\sin^2\varphi_T)^{1/2}} \approx a\left[1+\frac{e^2}{2}\sin^2\varphi_T\right]$$

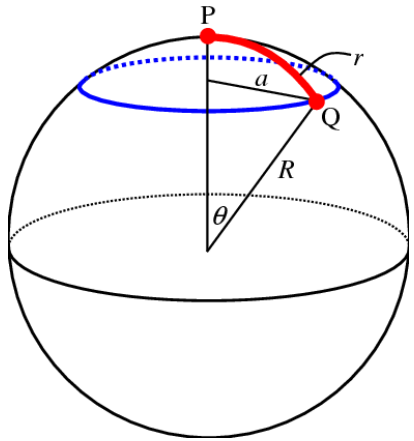


Radius of Curvature

➢ **Gaussian radius of Curvature R_G**

It is the radius of the best fitting sphere to the ellipsoid at any point.

$$R_G = \sqrt{R_M R_P} \approx a\left[1 - \frac{e^2}{2}\cos 2\varphi_T\right]$$



✓ These values are directly involved in calculating:

- **Change in latitude**
- **Change in longitude**

"Because the Earth is curved, linear aircraft velocity must be related to angular changes in latitude and longitude using the Earth's local radii of curvature".

$\varphi_T = F_T$

Radius of Curvature

The radii of curvature are important because they relate the horizontal components of velocity to angular coordinates, such as latitude and longitude.

$$\dot{\varphi}_T = \frac{V_{\text{north}}}{R_M + h}$$

$$\dot{\imath} \cos\varphi_T = \frac{V_{\text{east}}}{R_P + h}$$

h : the aircraft's altitude above the reference ellipsoid,

$\imath$: its longitude, measured positively east.

- For numerical work

a= 6378.137 km = 3443.918 nmi

- ✓ f = 1/298.2572
- ✓ $e^2 = f(2 - f)$

One nautical mile = 1852 meters = 6076.11549 ft

- ✓ The angle between the gravity vector and the normal to the ellipsoid, *the deflection of the vertical*, is commonly less than 10 seconds of arc and is rarely greater than 30 seconds of arc.

- ✓ The magnitude of gravity at sea level is **g = 978.049(1+0.00529 sin² φ_T) cm/sec²**

Coordinate Frames

- The aircraft's position, velocity, and attitude must be expressed relative to a coordinate frame.
- A coordinate frame is a reference system used to describe the position, velocity, or attitude of an aircraft.

❑ **This means we need:**

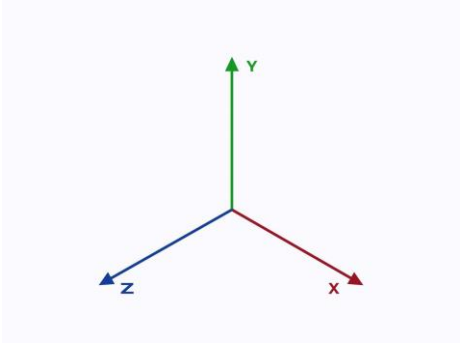
- An origin
- Coordinate axes
- Defined directions

so that we can determine **where the aircraft is and how it is moving**.

✓ There is no single coordinate frame that is ideal for every navigation calculation.

✓ The book presents **nine main types of coordinate frames** in this section.

Coordinate Frames



➢ **For example, in 2D, If we say: `(x, y) = (5, 3)`**
you know the position of the object **relative to this coordinate frame**.

➢ However, in aviation, the situation is more complex because:

- The aircraft moves in **3D**.
- The Earth rotates.
- The Earth is not flat.
- The aircraft itself changes its **orientation**.
- Sometimes we need a reference frame that is **fixed to the Earth**, while other times we need a reference frame that is **fixed relative to the stars**.

✓ Therefore, we need **different coordinate frames for different purposes**.

Coordinate Frames

⋆ **The Basic Idea**

At the same aircraft, We can describe its position:

1. **Relative to the center of the Earth >>> ECEF**
2. **Relative to the stars / inertial space >>> ECI**
3. **Using Latitude, Longitude, and Altitude >>> Geodetic coordinates**
4. **Relative to the aircraft itself >>> Body coordinates**

- The physical aircraft has only one position, but that position can be represented in different coordinate frames.

Coordinate Frames

1. Earth-Centered, Earth-Fixed - ECEF

❑ **Earth-Centered :** The origin is at the center of the Earth.

❑ **Earth-Fixed :** This means that the **coordinate system rotates with the Earth**.

In other words, the axes are **fixed relative to the Earth**.

For example, the axes are defined in relation to:

- The Earth's rotation axis
- The Greenwich/reference meridian

✓ ECEF is the basic coordinate frame used for navigation near the Earth, with its axes fixed relative to the Earth.

✓ **Why Is ECEF Useful?**

Aircraft navigation usually takes place near the Earth.

Therefore, we want to know: **Where is the aircraft relative to the Earth?** For this reason, ECEF is very useful for:

- Aircraft navigation
- Satellite positioning
- GPS calculations

✓ **An Important Point :** ECEF is **not an inertial frame**. Why?

Because the **Earth itself is rotating**. Therefore: **ECEF rotates with the Earth.**

Coordinate Frames

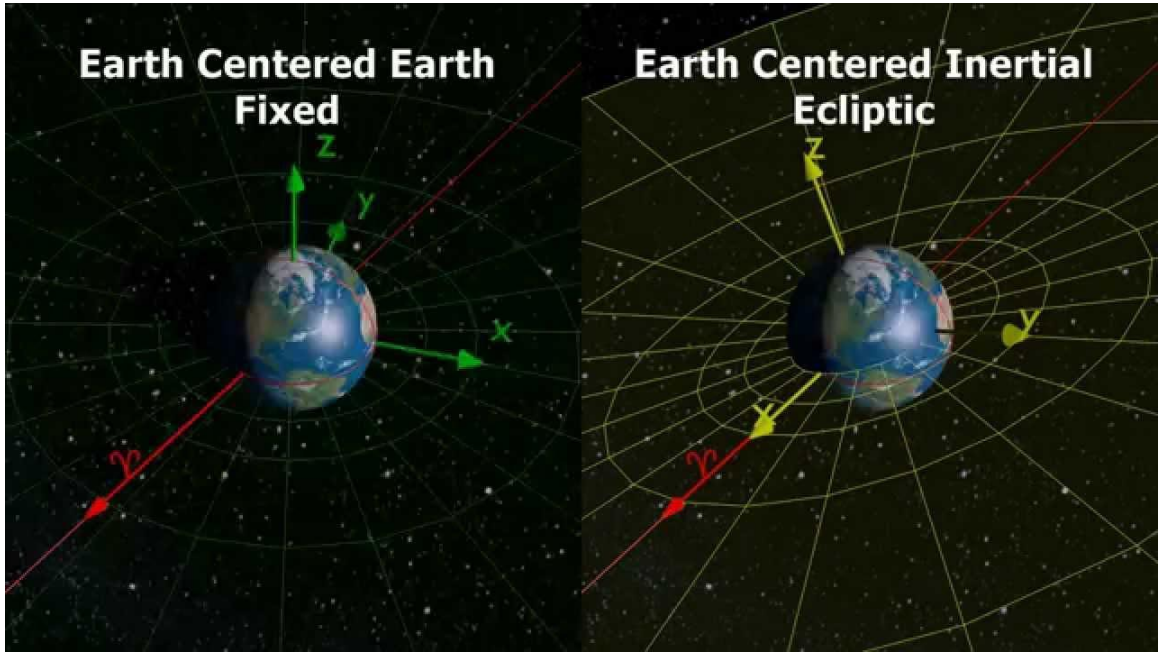
2. Earth-Centered Inertial — ECI

✓ **What Is the Difference from ECEF?**

Both coordinate frames are **Earth-Centered**, meaning that their **origin is located at the center of the Earth**.
However, the key difference is:
ECEF is Earth-Fixed and ECI is Inertial

✓ The **ECI coordinate system does not rotate with the Earth**.
In other words, it is considered **fixed relative to distant stars**.

✓ The **Earth rotates inside this coordinate system**.
✓ The ECI frame can be chosen so that one of its axes is **approximately aligned with the mean polar axis of the Earth**, providing a reference associated with the motion of the Earth relative to the stars.



Coordinate Frames

ECEF	ECI
Earth-Centered	Earth-Centered
Fixed to Earth	Inertial reference
Rotates with Earth	Does not rotate with Earth
Useful for Earth navigation	Useful for inertial/space reference
Greenwich/earth-fixed axes	Star/inertial reference

✓ ECEF rotates with the Earth, while ECI is treated as non-rotating relative to the distant stars.

Coordinate Frames

3. Geodetic Spherical Coordinates

These are the coordinates that are closest to the **coordinates we commonly use in everyday life and in aviation**. They consist of: **Latitude**, **Longitude, Altitude**

- ✓ The **geodetic coordinates are based on the normal to the reference ellipsoid** and are commonly used in **dead-reckoning and radio-navigation calculations**.

✓ **Example : If we say:**

Latitude = 30° N
Longitude = 31° E
Altitude = 10,000 ft

This represents the aircraft's position using geodetic coordinates.

➢ **Why Are They Called Geodetic?**

Because they are based on a **reference ellipsoid**, rather than a perfect sphere.

- ✓ Remember from the previous section: 🌍 **Earth ≠ a perfect sphere**
- ✓ **Earth → approximately an ellipsoid**
- ✓ Therefore, the latitude used in navigation is defined with respect to the reference ellipsoid.

Coordinate Frames

4. Geodetic Wander-Azimuth Coordinates

This is a coordinate system based on **geodetic coordinates**, but the **horizontal axes are allowed to "wander," or rotate, in a controlled manner**.

- ✓ These coordinates are **locally level with respect to the reference ellipsoid** and are particularly suitable for some **worldwide inertial navigation computations**.

➢ Why "Wander"?

Imagine an aircraft flying north.
If we keep certain axes fixed in a particular way, problems can occur, especially when the aircraft approaches the **polar regions**.
Therefore, we allow the **azimuth axis** to change or **"wander"** instead of keeping it fixed to a geographical direction that may become unsuitable.

- ✓ **The horizontal reference direction is allowed to rotate gradually to avoid problems associated with fixed directions, especially near the poles.**

Coordinate Frames

5. Direction Cosines

➢ What Are Direction Cosines?

Sometimes we have **Coordinate Frame A** and **Coordinate Frame B**
The same **vector** exists in both frames, but its components are different.
The vector is the same, but its **components change because the axes have changed**. Therefore, we need a way to transform a vector from **one coordinate frame to another**.

This is where the **Direction Cosine Matrix — DCM** comes in.
The **direction cosines** as a way to transform the coordinates of a vector from one coordinate system to another.

⋆ **The Main Equation of** transformation can be written as: $[V]_x = [C_{xy}][V]_y$

Direction Cosine Matrix will convert the Vector in frame y to **Vector in frame x**
So, the DCM tells us how to express the **same physical vector** using the axes of another coordinate frame.

➢ What Is Inside the Matrix?

The matrix contains the **cosines of the angles between the axes of the two coordinate systems**.

$$C = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix}$$

Each element, such as $\mathbf{C_{ij}}$, describes part of the **orientation relationship between the axes** of the two coordinate frames.

✓ The vector does not physically change. Only its representation changes because the coordinate axes are different.

Coordinate Frames

✈ Why Are Direction Cosines Important in Avionics?

An aircraft uses several different coordinate frames.

➢ **For example:**

Body Frame : Axes that are **fixed to the aircraft**.
Navigation Frame : Axes associated with the **local navigation reference frame**.
Earth Frame : Axes that are **fixed relative to the Earth**.
We need to perform transformations between these frames.

➢ **For example:** $V_{body} \rightarrow V_{navigation}$

This concept is fundamental to:

- Inertial Navigation Systems (INS)
- Attitude determination
- Flight control
- Sensor measurements

✓ **The vector does not change physically; only its components change when we describe it in a different coordinate frame.**

Coordinate Frames

6. Map-Grid Coordinates

These coordinates are based on **maps**. examples such as:

- **Lambert Conformal**
- **Transverse Mercator**
- **UTM (Universal Transverse Mercator)**

➢ Why Do We Need Map Coordinates?

Working directly with **Latitude + Longitude ,** may not always be convenient when working with a map.
Therefore, we can transform geographic coordinates into >> x, y on a map.

❑ **Example :** Instead of: **Latitude = 30° & Longitude = 31°**
we can use: **Easting = .. & Northing = ..**
This can be more convenient for certain applications.

➢ But There Is a Problem

We cannot transform the curved surface of the Earth onto a **flat map** without introducing some **distortion**.
Therefore, different **map projections** have been developed to minimize or control this distortion.

❑ **For example:**
Lambert Conformal >> Useful for some **aviation charts**.
Transverse Mercator >> Used in some **geographic coordinate systems**.

✓ These coordinate systems may not be suitable for some **long-range navigation calculations** because of the distortion introduced by map projections, particularly in **polar regions**.

Coordinate Frames

7. Geocentric Spherical Coordinates

Here we return to the spherical coordinate system.

➢ **Geocentric** means that the reference point is the **center of the Earth**.

Therefore, we have:

- **Radial distance**
- **Angular coordinates**

✓ The **radius vector** from the center of the Earth to the aircraft is the basis of this coordinate system.

✓ The **geocentric spherical coordinates** are used in some **short-range dead-reckoning systems** and in applications such as **spherical astronomy**.

➢ Geodetic vs. Geocentric

This is an important distinction that can sometimes cause confusion.

- **Geodetic** coordinates are based on a **reference ellipsoid**.
- **Geocentric :** the position and angular coordinates are defined relative to the **center of the Earth**.

✓ Because the Earth is an **ellipsoid** rather than a perfect sphere:

✓ Geodetic latitude ≠ Geocentric latitude

In other words, the two latitudes are measured using **different references**, so they are not necessarily the same.

Coordinate Frames

8. Transverse-Pole Spherical Coordinates

This is a **similar spherical coordinate system**, but instead of using the Earth's geographic pole as the reference, a **transverse pole** is selected.

- ✓ These coordinates are similar to **geocentric spherical coordinates**, but the poles are placed in **transverse positions** relative to the Earth's geographic pole.

➢ Why Do We Use It?

The basic idea is to choose a coordinate system that makes certain calculations more convenient for particular **regions or directions of motion**.

- ✓ This type of coordinate system has historically been used in some **inertial navigation** and **Doppler navigation systems**.

Coordinate Frames

9. Tangent-Plane Coordinates

Instead of dealing directly with the **Earth's curvature**, we use: **A local flat plane tangent to the Earth at the aircraft or reference point.**

At a particular point, we define a **plane that touches the Earth's surface** at that point.

➢ Why Is This Useful?

When performing navigation over a **small area**, we can approximate the Earth's surface as **flat**.
So instead of dealing with **A curved Earth ,** we use **A local flat plane**
This makes the calculations much simpler.

❑ Example :

If an aircraft is moving only a few **kilometers around an airport**, a **local tangent plane** can be used as an approximation.
However, if the aircraft is traveling from: **Cairo → New York :** we cannot treat the Earth as flat, because the Earth's curvature becomes significant.

Coordinate Frames

➢ **Why Do We Have Nine Coordinate Frames?**

Not because the book wants to make things complicated 😊
The main reason is that **no single coordinate system is ideal for every navigation calculation**.

✓ Each coordinate frame has a specific purpose and is useful for a particular type of **navigation problem or calculation**.
✓ So, depending on what we want to calculate—such as **position, velocity, attitude, or motion relative to the Earth**—we may choose a different coordinate frame.

✓ **Different coordinate frames are used because each one is designed to make a particular navigation calculation easier and more convenient.**

Coordinate Frames

Assume aircraft flying from Cairo to London.
Step 1 — Where is the aircraft relative to Earth?
Use: **ECEF / Geodetic coordinates**

Step 2 — We need inertial calculations
Use: **ECI**

Step 3 — We want local aircraft velocity
Use: **Local-level / tangent-plane type coordinates**

Step 4 — Sensor is mounted on aircraft
Its measurements are initially in: **Body coordinates**
Need to transformation to navigation frame. By using **Direction Cosine Matrix**

Step 5 — We want to display route on a map
Use: **Map-grid coordinates / digital chart coordinates**

"A coordinate frame is simply a reference system used to describe the position, velocity, and attitude of the aircraft. Different coordinate frames are used because different navigation problems require different references."

Participation

Scenario 1 – Aircraft Attitude

An aircraft is flying with a heading of 045◦045^\circ, pitch of 5◦5^\circ, and roll of 10◦10^\circ.

Questions:

1. Which coordinate frame is most suitable for describing the aircraft's roll, pitch, and yaw?
2. Why is this frame used rather than the NED frame?
3. What physical quantities are normally expressed in this frame?

Scenario 2 – Aircraft Position

An aircraft is flying over Cairo at a known latitude, longitude, and altitude.

Questions:

1. Which coordinate frame would you use to represent the aircraft's position relative to the Earth's surface?
2. What are the three position coordinates?
3. Why is this frame useful for aircraft navigation?

The nine frames can be treated as:

1. **Earth-Centered Inertial (ECI) frame**
2. **Earth-Centered Earth-Fixed (ECEF) frame**
3. **Local-level / Navigation frame (NED: North-East-Down)**
4. **Geodetic frame**
5. **Body frame**
6. **Vehicle frame**
7. **Platform / Computation frame**
8. **Wander-azimuth frame**
9. **Sensor / Instrument frame**

Participation

Scenario 3 – Local Navigation

An aircraft is flying north-east and descending toward an airport.

Its velocity is:

$$V_N = 150\,m/s, \qquad V_E = 100\,m/s, \qquad V_D = 5\,m/s$$

Questions:

- Which coordinate frame are these velocity components expressed in?
- What does each component represent?
- Why is the Down direction used instead of Up?

Scenario 4 – GPS Position Calculation

A GPS receiver determines the aircraft position using Cartesian coordinates:

$$X = 4.2 \times 10^6\,m$$

$$Y = 1.7 \times 10^6\,m$$

$$Z = 4.6 \times 10^6\,m$$

Questions:

- Which coordinate frame is most likely being used?
- Why is this frame useful for GPS?
- How could these coordinates eventually be converted into latitude, longitude, and altitude?

The nine frames can be treated as:

1. **Earth-Centered Inertial (ECI) frame**
2. **Earth-Centered Earth-Fixed (ECEF) frame**
3. **Local-level / Navigation frame (NED: North-East-Down)**
4. **Geodetic frame**
5. **Body frame**
6. **Vehicle frame**
7. **Platform / Computation frame**
8. **Wander-azimuth frame**
9. **Sensor / Instrument frame**

Dead-Reckoning Computations

- **Dead Reckoning (often called DR)** is the technique of calculating position from velocity measurements.
- It is the means of navigation in the absence of position fixes and consists of calculating the position (the zi-coordinates) of a vehicle by extrapolating (integrating) estimated or measured ground speed.

Dead Reckoning (DR): **Determining the current position from a previously known position and measurements of motion.**

- If I know **Where I started?** and I know **How fast I moved and in which direction ?** I can calculate **Where I am now.**
- **Basic Flat-Earth Dead Reckoning :** In its simplest form, neglecting wind, dead reckoning can calculate the position of a vehicle on the surface of a flat Earth from measurements of ground speed and true heading.

$$V_{north} = V_g \cos W_T \quad , \quad y - y_0 = \int_0^t V_{north}\, dt$$

$$V_{east} = V_g \sin W_T \quad , \quad x - x_0 = \int_0^t V_{east}\, dt$$

- V_g : ground speed
- W_T : true heading
- x-xo and y-yo : the east and north distances traveled during the measurement interval

$$Position = \int Velocity\, dt$$

Dead-Reckoning Computations

- **Example :**

$$x_0 = 0, \quad y_0 = 0$$

and travels East at: $V = 100\,knots$

$$t = 2\,hours$$

$$Distance = 100 \times 2 = 200\,nm$$

$$x = 200\,nm$$

$$y = 0$$

- **The Problem with Dead Reckoning** If there is a small error in (**Speed, Heading, Drift, Wind)** this error **accumulates over time**.
- As the flight continues, the estimated position moves farther and farther away from the aircraft's **actual position**.
- ✓ Therefore: **Dead-reckoning position errors generally grow with time.**

Dead-Reckoning Computations

- The aircraft has **Airspeed.** The aircraft's **velocity relative to the air**.
- However, aircraft navigation needs to know **Ground Velocity**

The aircraft's **velocity relative to the ground**.

- The **wind** acts as an additional velocity vector.

$$\vec{V}_{ground} = \vec{V}_{air} + \vec{V}_{wind}$$

- In the presence of a **crosswind**, the aircraft's velocity vector relative to the air is not necessarily in the same direction as the **ground track**.
- This means we have several vectors:

- **True Airspeed (TAS)** → the aircraft's velocity relative to the air.
- **Wind velocity** → the motion of the air mass itself.
- **Ground velocity** → the aircraft's velocity relative to the ground.

✓ The **geometry of dead reckoning**, including the **wind vector, drift angle, heading, and ground track**.

Dead-Reckoning Computations

➢ What does a Crosswind do?

Imagine that the aircraft wants to fly **north,** If there is a **crosswind** from the side, the wind pushes the aircraft sideways.

- Therefore: **Aircraft heading ≠ Ground-track direction**
- The angle between them is called the **Drift Angle (δ)**

➢ Drift Angle

The **drift angle** is the difference between the **Aircraft heading** and the **Ground track direction.**
This difference is caused by the **wind**.

- **For example,** suppose the aircraft's heading is 90° , but the wind pushes the aircraft toward the north.

As a result, the actual **ground track** may become 85° or another value, depending on the **wind vector**.

➢ Doppler Radar : **Doppler radar** can measure **Ground speed** and **Drift angle**

✓ The **Dead-Reckoning computer** can use information obtained from the **Doppler radar**.
✓ The Doppler radar can measure:

- **Ground speed** V_g
- **Drift angle** δ

$$V_{\text{north}} = V_g \cos(\psi_T + \delta)$$
$$V_{\text{east}} = V_g \sin(\psi_T + \delta)$$

✓ This is very useful because the computer gets direct information about the aircraft's **motion relative to the ground**, rather than relying only on **airspeed** and the **predicted wind**.

Dead-Reckoning Computations

$$V_{north} = V_{TAS}\cos(\theta - \alpha)\cos(\psi_T + \beta) + V_{wind-north}$$

$$V_{east} = V_{TAS}\cos(\theta - \alpha)\sin(\psi_T + \beta) + V_{wind-east}$$

V_{TAS} **True Airspeed** The aircraft's speed **relative to the air mass.**

θ **Pitch Angle** The angle of the aircraft relative to the **horizontal plane.**

α **Angle of Attack** The angle between the aircraft's **reference line** and the **relative airflow.**

β **Sideslip Angle** The angle describing the aircraft's **lateral deviation relative to the direction of the airflow.**

ψ_T **True Track Angle**

The direction of the aircraft's **motion relative to the ground**, measured from the reference direction used for navigation.

$V_{\text{wind-north}}$ The **North component of the wind velocity.**

$V_{\text{wind-east}}$ The **East component of the wind velocity.**

Dead-Reckoning Computations

➢ **What happens after we obtain V_{north} and V_{east}?**

This brings us to a very important concept in **Dead Reckoning**.
On a **flat Earth**, we can simply integrate velocity with respect to time:

$$\Delta North = \int V_{north}\ dt$$

$$\Delta East = \int V_{east}\ dt$$

In other words: **Velocity → integrate over time → Distance traveled**
Then, we use the distance and direction to determine the aircraft's **new position**.

🌍 **But the Earth is not flat**

This is where the concept connects to what we discussed earlier about **Earth geometry**.
On the real Earth, we cannot simply work with : North distance + East distance
Instead, position is expressed using **angular coordinates**, such as: **Latitude** & **Longitude**

➢ Therefore, we need to convert the **linear velocities** into **rates of change of latitude and longitude**.

$$V_{north} \rightarrow \dot{\phi}$$
$$V_{east} \rightarrow \dot{\lambda}$$

✓ This is where the **radii of curvature**, R_M and R_P, that we discussed earlier become important.

Dead-Reckoning Computations

- **Airspeed tells us how the aircraft moves relative to the air, but navigation needs to know how the aircraft moves relative to the ground.**
  - Therefore, **wind and drift are essential components of dead-reckoning calculations.**

Positioning

How can we determine the aircraft's actual position?

1. Radio Fixes : There are **five basic airborne radio measurements**.

I. Bearing

Bearing = the angle of arrival of a radio signal.

In other words: **From which direction is the signal coming?**

II. Phase

The receiver measures the **Phase difference** between continuous-wave signals.

III. Time Difference

The receiver measures the difference between the arrival times of signals from different stations.

$$\Delta t = t_1 - t_2$$

This concept is important in systems such as **hyperbolic navigation**.

IV. Two-Way Range

The aircraft: **transmits a signal → station → signal returns to the aircraft**

The **round-trip time** is measured.

$$\boxed{R = \frac{c\Delta t}{2}}$$

- c= speed of light
- Δt= round-trip time

V. One-Way Range

Here, the receiver measures the **Time of arrival (TOA) of a signal**

However, the system must have **synchronized or known timing**.

This concept is very important later when studying **GPS pseudorange.**

Positioning

2. Line-of-Sight Distance Measurement

Here, we want to calculate the distance between the **Aircraft** and the **Radio station** on or around the **Earth's surface**.

We have the line-of-sight vector: $\vec{R}_i - \vec{R}_0$

❑ Using geometry, we can calculate: **Distance** & **Elevation** & **Azimuth**

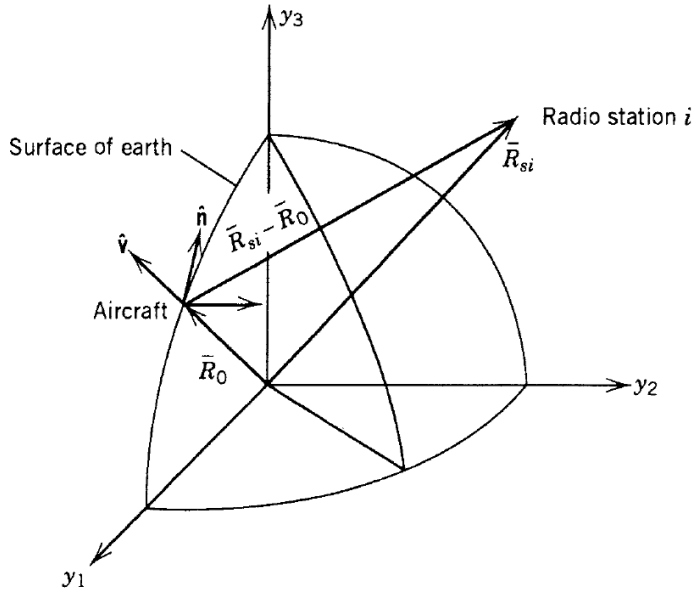


Figure 2.5 Light-of-sight distance.

Positioning

➢ **Elevation Angle**

If we have an **aircraft** and a **radio station**:

The **elevation angle** is the angle between the: **horizontal plane** and the **line of sight to the station.**

➢ **Azimuth Angle**

Azimuth is the horizontal direction of the **line of sight** measured relative to a **reference direction**.

❑ For example:

The station is **40° clockwise from North**.

Therefore:

Az=40∘

Positioning

- The equations based on **vector geometry** and **Earth-centered coordinates**.

Figure 2.5 shows an aircraft near the surface of the Earth at $\mathbf{R}_0$ and a radio station that may be near the surface or in space, at $\mathbf{R}_{si}$. The slant range, $|\mathbf{R}_{si} - \mathbf{R}_0|$, from the aircraft to the station could be measured by one-way or two-way ranging. If $\hat{\boldsymbol{v}}$ is the unit local vertical vector at the aircraft, the elevation angle of the line of sight to the radio station is

$$\sin E = \frac{\hat{\mathbf{v}} \cdot (\mathbf{R}_{si} - \mathbf{R}_0)}{|\mathbf{R}_{si} - \mathbf{R}_0|} \tag{2.14}$$

If $\hat{\mathbf{n}}$ is the unit north-pointing horizontal vector at the aircraft, the azimuth angle of the line of sight is

$$\sin Az = \frac{\hat{\mathbf{v}} \times (\mathbf{R}_{si} - \mathbf{R}_0)}{|\mathbf{R}_{si} - \mathbf{R}_0| \cos E} \times \hat{\mathbf{n}} \tag{2.15}$$

These vector equations can be resolved into any coordinate frame.

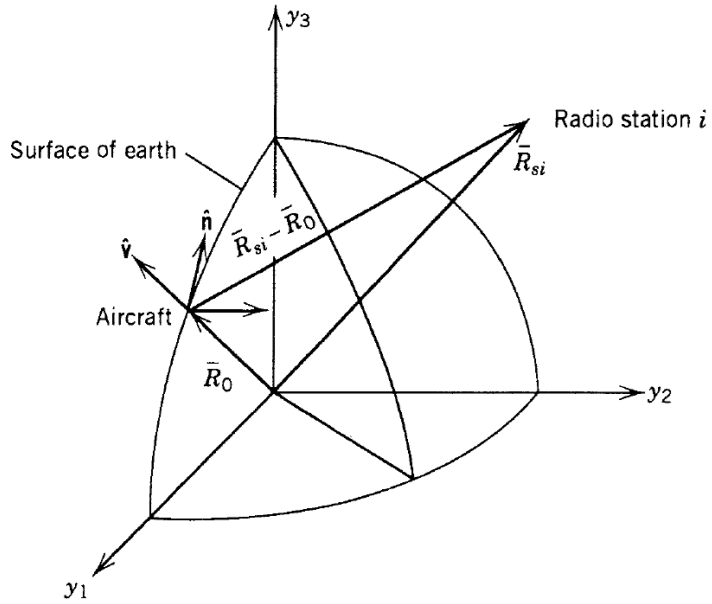


Figure 2.5 Light-of-sight distance.

Positioning

3 Ground-Wave One-Way Ranging

Here, the signals propagate along the **curved surface of the Earth**.
The basic idea is that the aircraft measures the **Arrival time of the signals,** From these measurements, the navigation system can determine the **aircraft's position**.

4 Ground-Wave Time-Differencing

Here, instead of measuring the **absolute arrival time**, we measure the **Difference between the arrival times of signals from two stations.**

$$\boxed{\Delta t = t_A - t_B}$$

- t_A= arrival time of the signal from station A
- t_B= arrival time of the signal from station B
- Δt= difference in arrival times

➢ **How does the navigation computer determine the position?**

It uses an **iterative positioning process**:

1. Assumes an aircraft position.
2. Calculates the predicted signal arrival times for that position.
3. Measures the actual time difference between the two signals.
4. Compares the measured and predicted values.
5. Calculates the error (residual).
6. Updates the estimated aircraft position.
7. Repeats the process until the residual becomes sufficiently small.

➢ **In simple words:** The computer keeps asking: **"If the aircraft were here, what time difference should I expect?"**
Then it compares this with: **"What time difference did I actually measure?"**
It adjusts the aircraft position and repeats the calculation until the predicted and measured values are close enough.
So, this is essentially an: Iterative Positioning Process
This method is particularly useful because **time differences can provide positional information without requiring an absolute clock reference**.

Terrain-Matching Navigation

➢ **Can the aircraft determine its position by looking at the terrain below it?**

Yes: The basic idea is to compare the **measured terrain** with a **stored terrain map/profile**.

1. Stored Terrain Profile

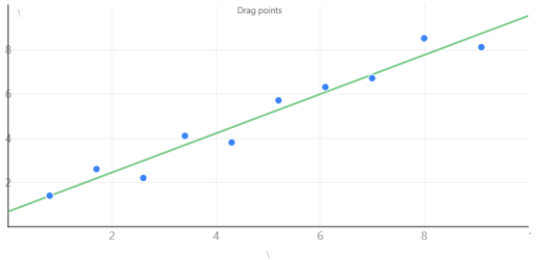
The navigation system has a **digital terrain database** stored in memory.
It contains information about the elevation of the terrain along different locations.

2. Actual Terrain Profile

The aircraft uses sensors, typically a **radar altimeter**, to measure the terrain below it.
As the aircraft moves, it obtains a sequence of terrain elevations **Actual terrain profile**

3. Correlation / Matching

The system compares **Measured terrain profile** with **Stored terrain profile** using **correlation / pattern matching**.
If there is a strong match, the system can determine: **"The terrain I'm measuring looks like the terrain stored for this particular location.“** Therefore, the aircraft can estimate its **position**.



The aircraft does not need to "see" the ground visually. Instead: **It measures the terrain elevation and compares the measured pattern with a stored terrain map.**

✓ So the navigation chain is: **Terrain → Measurement → Profile → Correlation/Matching → Position Estimate**

Course Computation

- After determining **Where am I?**
- The next question is **Where should I go?**

To answer this, the navigation system needs to determine:

I. **Range and Bearing Calculation**
II. **Direct Steering**
III. **Airway Steering**
IV. **Area Navigation (RNAV)**

- **What is Course Computation?**

The navigation system have current aircraft position and destination and waypoint.

The navigation computer calculate :

- **Range** → How far is the destination?
- **Bearing** → In which direction is the destination?
- **Course/Track** → Which direction should the aircraft fly?
- **Steering command** → How should the aircraft be turned?

I. Range and Bearing Calculation

If we know the: **Current position** and the **Destination**
we want to calculate two important quantities:

1. Range : the distance from the aircraft to the destination.
For a flat-Earth approximation:

$$D = \sqrt{(x - x_2)^2 + (y - y_2)^2}$$

where the difference in the two coordinates gives the horizontal displacement between the aircraft and the destination.

2. Bearing : the direction from the aircraft to the destination.

In other words: **In which direction should the aircraft travel to reach the destination?**

- ✓ The navigation computer uses the aircraft's position and the destination coordinates to calculate the required **range and bearing**. which provides smoother guidance by considering the geometry and rate of change of the line of sight to the target.

I. Range and Bearing Calculation

➢ True Bearing vs. Relative Bearing

This is a **very important concept in aircraft navigation**, because the reference direction changes depending on what we are measuring from.

1. True Bearing B_T

True Bearing is the angle measured from **True North** It tells us the direction of the destination relative to geographic North.
Therefore:
Ex: $B_T = 60°$ This means that the destination is located **60° clockwise from True North**.

2. Relative Bearing B_R

Relative Bearing is measured from the **Longitudinal axis of the aircraft**
In simple words, we measure the angle starting from the **aircraft's face**.
The relationship is: $\boxed{B_R = B_T - \psi_T}$

- B_R= **Relative Bearing**
- B_T= **True Bearing**
- ψ_T= aircraft **heading/track reference** used in this context

range D $\quad D = [(x - x_t)^2 + (y - y_t)^2]^{1/2}$

bearing B_T to the target $\quad B_T = \tan^{-1} \dfrac{x - x_t}{y - y_t}$

➢ **Example :** Suppose Aircraft heading $\psi_T = 30°$, Target true bearing $B_T = 70°$, Then:

$$B_R = 70° - 30° = 40°$$

So, the target is **40° to the right of the aircraft's face.**

✓ The important point is that **True Bearing is measured from North**, while **Relative Bearing is measured from the aircraft's own longitudinal axis**.

I. Range and Bearing Calculation

3. Relative Track

Relative Bearing and **Relative Track**.
Relative Bearing Measured relative to **Aircraft longitudinal axis / heading**

Relative Track Measured relative to **Aircraft ground track**

$$\boxed{T_R = B_T - T_T}$$

- T_R= **Relative Track**
- B_T= **True Bearing to the target**
- T_T= **True Ground Track**

Bearing → Where is the target?
Heading → Where is the face pointing?
Track → Where is the aircraft actually moving?

➢ **Why is this distinction important?**

An aircraft can have a **heading** that is different from its **ground track**, especially when **wind** is present.

Heading → where the aircraft's face is pointing.
Ground Track → the actual direction in which the aircraft is moving over the ground.

✓ Therefore, when wind causes **drift**, the **Relative Bearing** and **Relative Track** can be different.

I. Range and Bearing Calculation

➢ The **range and bearing** can be calculated using either:

1. **Plane triangle**
2. **Spherical triangle**

1. Plane-Triangle Solution

When the **flat-Earth approximation** is acceptable, we can use the **plane-triangle solution**.

❑ **Range Calculation :** The distance between the aircraft and the target can be calculated using the **Pythagorean theorem**:

$$\boxed{D = \sqrt{(x - x_T)^2 + (y - y_T)^2}}$$

- D= **range to the target**
- x, y= **current aircraft coordinates**
- x_T, y_T= **target coordinates**

❑ **Example :**

Aircraft position $(x, y) = (10, 20)$ and Target position $(x_T, y_T) = (40, 60)$

First, calculate the coordinate differences:

$$\Delta x = 40 - 10 = 30$$
$$\Delta y = 60 - 20 = 40$$

$$a^2 + b^2 = c^2$$

Therefore:

$$D = \sqrt{30^2 + 40^2} \quad \gg \quad D = \sqrt{900 + 1600}$$
$$\boxed{D = 50}$$

So, the target is **50 distance units away** from the aircraft.

I. Range and Bearing Calculation

❑ Bearing Calculation

For the flat-Earth coordinate system used here, the bearing can be obtained from the coordinate differences.

$$B_T = \tan^{-1}\left(\frac{x - x_T}{y - y_T}\right)$$

🌍 Why Can't We Always Use the Plane Triangle?

The main reason is **The Earth is curved.**

- ✓ For relatively short distances, we can approximate the Earth as flat, But as the distance increases, the curvature of the Earth becomes significant.
- ✓ Therefore: **The flat-Earth approximation becomes less accurate as the range increases.**
- ✓ For long-range navigation, we use a **spherical-triangle solution** instead.

❖ Example of the Approximation Error

$D = 1000\ nmi$, with $B_T = 45°$, and $\Phi = 45°$

Under these conditions, the error introduced by the flat-Earth approximation can be approximately $\boxed{36\ nmi}$

That's a **very significant error** for aircraft navigation.

However, for a distance of approximately: $D \leq 100\ nmi$

the book indicates that treating the Earth as flat can produce an error of about: $\boxed{0.3\ nmi}$

which may be acceptable depending on the navigation application.

✈ Key Points :

Short distance: Plane Triangle → acceptable approximation

Long distance: Spherical Triangle → account for Earth curvature

So the navigation computer has to choose the appropriate mathematical model depending on the **distance and required accuracy**.

I. Range and Bearing Calculation

2. Spherical-Triangle Solution

When the distance is large, we must account for the **curvature of the Earth**, so we use **spherical geometry** instead of flat-Earth geometry.

❑ Range Calculation on the sphere :

$$\cos\frac{D}{R_G} = \sin\Phi\sin\Phi_T + \cos\Phi\cos\Phi_T\cos(\lambda - \lambda_T)$$

- D= **range to the target**
- R_G= **Gaussian radius of curvature**
- Φ= **aircraft latitude**
- Φ_T= **target latitude**
- λ= **aircraft longitude**
- λ_T= **target longitude**

❑ What does the equation mean?

Instead of assuming that the aircraft and target are on a flat plane, we consider the actual curvature of the Earth:

✓ Therefore, Drepresents the **distance along the curved-Earth geometry**, rather than a straight-line distance calculated using a flat coordinate system.

❑ Bearing calculation on the Sphere

For calculating the target's **True Bearing**:

$$\sin B_T = \frac{\cos\Phi_T}{\sin(D/R_G)}\sin(\lambda - \lambda_T)$$

This equation is used together with the spherical range calculation to determine the **direction of the target**.

✓ So, with spherical navigation:

Latitude + Longitude → Range + Bearing

I. Range and Bearing Calculation

Situation	Method
Short distance	Plane triangle
Small area	Flat-Earth approximation
Long distance	Spherical triangle
Global navigation	Earth curvature must be considered

- ✓ **Short distance → Earth can be treated as flat.**
- ✓ **Long distance → Earth must be treated as curved.**

I. Range and Bearing Calculation

❖ Now we have determined **Range** and **Bearing.**

- So the navigation problem changes from **Where am I?** to **How do I get there?**

The navigation system must now generate appropriate **steering commands**.

➢ ***There are two main approaches:***

1. Direct Steering

The aircraft is guided **Directly toward the destination.**

Aircraft ✈ ——————→ Destination

2. Airway / Route Steering

The aircraft follows a **Preplanned airway or route**

Aircraft ✈ → Waypoint → Waypoint → Destination

And this leads us to the next important topics: Direct Steering and Airway Steering / RNAV

✓ **Position determination → Course computation → Steering the aircraft**.

1. Direct Steering

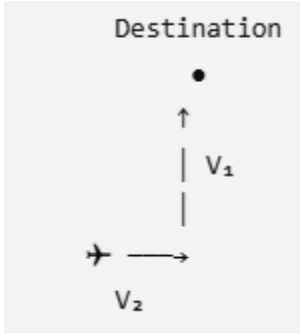
The meaning of **Direct Steering** is: **Steer directly from the present aircraft position toward the destination.**
In other words, the aircraft attempts to fly **directly to the target,** rather than following an intermediate waypoint.

➢ **How does the system know how to steer?**
The **steering computer** analyzes the aircraft's **ground velocity** and resolves it into two components relative to the **line of sight to the destination**.

❖ **V1 : Ground-speed component along the line of sight to the destination.**
It answers: **How fast am I moving toward the target?**

❖ **V2 : Ground-speed component perpendicular to the line of sight.**
It answers: **How much am I moving sideways relative to the desired direction?**

✓ V_1 is **parallel to the line of sight**, while V_2 is **perpendicular to it**.



➢ If: $\boxed{V_2 = 0}$
That's mean , there is no velocity component perpendicular to the line of sight.
Therefore, the aircraft's velocity is directed **straight toward the target**.

❑ **Think of it this way:**
- $V_1 \rightarrow$ **Am I moving toward the target?**
- $V_2 \rightarrow$ **Am I moving sideways away from the desired direction?**

✓ The navigation system calculates the required **steering direction** and, depending on the guidance system, the required **bank angle** to change the aircraft's direction.

1. Direct Steering

➢ **Bank Angle Command** means : **The angle at which the aircraft is commanded to tilt (bank) its wings to make a turn.**

❑ ***For example:***

Bank angle = 0° → aircraft flies straight.
Bank angle = 10° → aircraft tilts 10° and starts turning.
Bank angle = 20° → aircraft turns more strongly.

In **Direct Steering** ϕ_c is the **commanded bank angle**.

$$\phi_c = K_1 V_2 + K_2 \dot{V}_2$$

- ϕ_c= commanded bank angle
- V_2= lateral velocity component
- $\dot{V}_2$= rate of change of the lateral velocity
- K_1, K_2= control constants

➢ **What does this mean physically?**

First term $K_1 V_2$ This means ; **The larger the lateral component, the larger the correction required.**
For example: $V_2 = 10$ means there is a relatively large lateral component, so the system requests a **larger steering correction**.
But if: $V_2 = 1$ the aircraft is much closer to the desired direction, so the required correction is **smaller**.

Second term $K_2 \dot{V}_2$ This provides a form of **anticipation**.
The system does not only ask "How large is the error?“ , It also asks: **"How quickly is the error changing?"**
This allows the system to begin correcting before the lateral motion becomes excessive.

1. Direct Steering

➢ **"How much should I bank the aircraft to bring it toward the desired direction?"**

If the aircraft is moving significantly sideways relative to the destination $V_2 = 10$ → computer may command a larger bank angle, e.g. **15°**.
If $V_2 = 1$ → only a small correction is needed, e.g. **3°**.

✓ **Bank Angle Command = the turn/tilt instruction sent by the navigation or flight-control system to the aircraft.**

➢ Why is the Bank Angle Limited?

The calculated bank angle is normally subject to a maximum limit.

❑ For example: $\boxed{\phi_{\max} = 15^\circ}$ Suppose the computer calculates: $\phi_c = 35^\circ$
but the allowed maximum is 15°. Then: Actual command = 15°
This prevents the guidance system from commanding unnecessarily aggressive maneuvers.

1. Direct Steering

➢ Problem with Direct Steering

As the aircraft gets close to the destination, the required steering direction can change rapidly.

Therefore: **The steering signal can fluctuate excessively near the target.**
This can result in unnecessary or excessive changes in aircraft heading.

✓ To reduce this problem, the book introduces **proportional navigation**, which provides smoother guidance by considering the geometry and rate of change of the line of sight to the target.

1. Direct Steering

➢ Great-Circle Navigation

For **short distances**, we can approximate the Earth as a flat surface.
However, for **long distances**, the curvature of the Earth becomes important.
Therefore, simple plane geometry is no longer sufficiently accurate.

✓ The important principle is: **The shortest route between two points on a sphere is a great-circle route.**
Therefore, long-range aircraft navigation often uses **great-circle calculations** to determine the appropriate route.
The great-circle path represents the shortest path over the spherical Earth.

2. Airway Steering

In **airway steering**, the aircraft does **not** simply point directly toward the final destination.
Instead, it follows a **Predefined airway or flight path.**

❑ **For example:**

Airport ✈ Waypoint 1 ✈ Waypoint 2 ✈ Waypoint 3 ✈ Destination

The navigation system continuously calculates:

- **Desired course**
- **Lateral deviation**
- **Steering command**

The aircraft is then guided so that it remains close to the desired airway.

For long-distance navigation, the airway can be defined using a **great-circle path**.

Course Computation

➢ Direct Steering vs. Airway Steering

- **Direct Steering**

Aircraft → Destination, The aircraft is continuously steered toward the destination.

- **Airway Steering**

Aircraft ✈ Airway ✈ Destination

Here, the aircraft does **not** simply point directly toward the destination. Instead, it must **Stay on a predefined route.**

The navigation system must therefore determine whether the aircraft is **on the desired route or deviating away from it**.

How Does Airway Steering Work?

The steering algorithm performs the following steps:

1. Define a desired great-circle path
The navigation system first determines the desired route between the relevant waypoints.
For long-distance navigation, this route can be a **great-circle path**.

2. Determine the aircraft's lateral deviation
The system calculates how far the aircraft is from the desired airway, and is represented by (L).

Lateral Deviation : The perpendicular distance between the aircraft and the desired path is called: **Lateral deviation (L)**

- ✓ The navigation system attempts to reduce this deviation and bring the aircraft back toward the desired path.

Lateral deviation = distance from the desired path

How Does Airway Steering Work?

3. Generate a bank-angle command

Based on the lateral deviation and its behavior, the navigation system calculates the required bank angle:
This tells the aircraft how strongly it should turn toward the desired airway.

$$\boxed{\phi_c = K_1 L + K_2 \dot{L} + K_3 \int L\,dt}$$

- $K_1 L$: If the aircraft is far from the airway $L \uparrow$ Therefore, the correction ↑
- $K_2 \dot{L}$: It looks at: **How fast is the lateral deviation changing?** It provides an anticipatory action.

Integral Term $K_3 \int L\,dt$

If there is a **constant wind** that continuously pushes the aircraft away from the airway, the proportional correction alone may leave a **steady-state displacement**.
The integral term accumulates over time and continues adding correction until: $\boxed{L \to 0}$

➢ **Bank Angle Limiting**

As with Direct Steering:
If the aircraft is very far from the airway, we do not want $\phi_c = 40°$

$$\boxed{\phi_c \leq \phi_{max}}$$

This prevents the steering commands from becoming excessive.

How Does Airway Steering Work?

4. Aircraft turns toward the desired path

The aircraft responds to the bank-angle command and moves back toward the desired airway.
The objective is:

That means the aircraft is brought back onto the desired route.

5. Select the next waypoint

When the aircraft reaches the current waypoint, the navigation computer selects the next waypoint.

6. Calculate a new desired track

The navigation system then calculates the desired track toward the new waypoint and repeats the process.
Therefore, the aircraft can continuously follow a sequence of connected waypoints.

Navigation Errors

➢ **How accurate is our navigation solution?**

The **navigation error** is defined as: $\boxed{\text{Navigation Error} = \text{Estimated Position} - \text{True Position}}$

In other words, it represents the difference between **where the navigation system thinks the aircraft is** and **where the aircraft actually is**.

➢ The book identifies five major sources of navigation errors:

1. Navigation sensor errors
2. Computer errors
3. Data-entry errors
4. Display errors
5. Flight technical error

Navigation Errors

➢ Navigation Sensor Error

For example, suppose a sensor measures 100.5 , while the true value is 100.0

Then : $Error = 100.5 - 100.0 = +0.5$ So the sensor has an error of **+0.5**.

The important point is that sensor errors are not necessarily constant. Different measurements may produce different errors.

➢ Statistical Description of Navigation Errors

not simply describe an error by saying **Error = 0.5**

because the error can change from one measurement to another.

❑ Therefore, we use statistical parameters to describe the behavior of the errors.

- **Mean**
- **Standard deviation**
- **Variance**
- **Correlation**

These parameters help us understand not only the size of the error, but also **how the errors are distributed**.

Navigation Errors

1. Mean : Suppose we have several error measurements: $e_1, e_2, \dots, e_N$

The mean error is:

$$\bar{e} = \frac{\sum e_i}{N}$$

The mean tells us the **average error** over many measurements.

- **Example :** Suppose:

$$e_1 = +1$$
$$e_2 = -1$$
$$e_3 = +2$$
$$e_4 = -2$$

$$\bar{e} = \frac{1 - 1 + 2 - 2}{4} = 0 \qquad \boxed{\bar{e} = 0}$$

This means that, **on average**, there is no systematic offset in these measurements. However, this does **not** mean that every individual measurement is accurate. The individual errors can still be large.

2. Standard Deviation : The standard deviation is represented by σ , It describes the **spread of the errors around their mean**.

Standard deviation tells us how much the errors vary from one measurement to another.

- A small: σ means the measurements are relatively close together.
- A large: σ means the measurements are more widely spread.

$$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2}$$

Small σ → small spread → lower uncertainty

Large σ → large spread → greater uncertainty

✓ So, in navigation systems, standard deviation is an important indicator of **navigation accuracy and uncertainty**.

Navigation Errors

3. Variance

Variance is closely related to standard deviation. It measures the average squared deviation of the measurements from their mean.

$$\boxed{\sigma^2 = \text{Variance}}$$

$$\boxed{\sigma = \sqrt{\text{Variance}}}$$

So if the variance increases, the standard deviation also increases.

4. Correlation : Now consider that navigation errors can occur in different directions.

❑ For example: Error in the **x-direction** , Error in the **y-direction** .

These two errors may be:

Independent The error in x does not depend on the error in y.

or:

Correlated : There is some relationship between the x-error and y-error.

For example, if x-error tends to increase when y-error increases, the two errors are positively correlated.

- ✓ This is important because the relationship between the errors affects the **shape and orientation of the overall error distribution**.
- ✓ For two-dimensional navigation errors, the book discusses the use of **statistics and principal axes** to characterize the error distribution.

Navigation Errors

➢ CEP — Circular Error Probable

One of the important concepts in navigation accuracy is: **Circular Error Probable (CEP)**

CEP represents a radius around the estimated position such that approximately **50% of the navigation errors fall within that circle**.
The circle represents the region within which approximately half of the position errors are expected to occur.

$$\boxed{CEP \approx 0.59(\sigma_x + \sigma_y)}$$

CEP is therefore a **two-dimensional measure of position accuracy**.

✓ A smaller CEP generally corresponds to a tighter concentration of position errors around the true position.

➢ 2DRMS

The term **2DRMS** is commonly used as a measure of **two-dimensional navigation accuracy**.

$$\boxed{2drms = 2\sqrt{\sigma_x^2 + \sigma_y^2}}$$

- σx = standard deviation of the error in the x-direction
- σy = standard deviation of the error in the y-direction

❖ The important difference is:

CEP → radius associated with approximately 50% of errors

2DRMS → another statistical measure used to describe 2D position accuracy

✓ So they are **different accuracy measures** and should not be treated as interchangeable.

Navigation Errors

➢ 3D Navigation Accuracy

So far, we considered two-dimensional position:

- x → horizontal error
- y → horizontal error

But aircraft navigation is actually three-dimensional. Therefore, we also need to consider the vertical error:

z → vertical error

and its standard deviation: σ_z

The 3D version becomes:

$$2drms_{3D} = 2\sqrt{\sigma_x^2 + \sigma_y^2 + \sigma_z^2}$$

So now the navigation accuracy considers errors in all three dimensions.

In other words:

- **2D navigation accuracy** $x + y$
- **3D navigation accuracy** $x + y + z$

✓ This gives us a more complete description of the aircraft's overall **three-dimensional navigation performance**.

Navigation Errors

➢ GDOP : Geometric Dilution of Precision

This is a **very important concept for GPS** .

- GDOP is not only related to sensor accuracy. Rather, it is related to **The geometry of the navigation measurements.**
- GDOP relates **measurement error to the resulting position error**, and that the **geometry of the measurements affects navigation accuracy**.

- **Geometric dilution of precision (GDOP) relates ranging errors to the dispersion in measured position.**
- **If three range measurements are made in orthogonal directions, the standard deviations in the aircraft's position error are the same as those of the three range sensors.**

However, if the range measurements are nonorthogonal or there are more than three measurements, the aircraft's position error can be slightly smaller or much larger than the error in each range.

❑ **HDOP Horizontal Dilution of Precision :** It is concerned with **horizontal position accuracy**.

❑ **TDOP Time Dilution of Precision :** It is concerned with the effect of geometry on the **time estimate**.

Navigation Errors

If the variances in ranging errors to each station are equal, σ_R^2, and if the uncorrelated variances of aircraft position are σ_x^2, σ_y^2 and σ_z^2 in locally level coordinates, then, by definition, the position dilution of precision is

$$(\text{PDOP})^2 = \frac{\sigma_x^2 + \sigma_y^2 + \sigma_z^2}{\sigma_R^2}$$

and the horizontal dilution of precision (HDOP) is

$$(\text{HDOP})^2 = \frac{\sigma_x^2 + \sigma_y^2}{\sigma_R^2}$$

In pseudoranging systems, the GDOP is

$$(\text{GDOP})^2 = (\text{PDOP})^2 + (\text{TDOP})^2$$

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Thank You!

See You next Lectures!!

Any Question?

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